



Linear-drive amplifier-less 112 Gbit/s PAM4 operation of a silicon-organic hybrid (SOH) Mach-Zehnder modulator at 265 mV_{pp}

S. SINGER,^{1,2,†,*} A. SCHWARZENBERGER,^{1,2,†} C. ESCHENBAUM,^{1,2}
M. MARTENS,^{2,3} A. MERTENS,³ G. DAGHER,⁴ L. VALENZIANO,⁵ S.
SARWAR,^{6,7} P. ERK,¹ H. Kholeif,¹ A. KOTZ,¹  T. ZWICK,⁵ S.
BRÄSE,^{6,7} W. FREUDE,¹  S. RANDEL,¹ AND C. KOOS^{1,2,3,8} 

¹Institute of Photonics and Quantum Electronics (IPQ), Karlsruhe Institute of Technology (KIT),
Engesserstr. 5, 76131 Karlsruhe, Germany

²SilOriX GmbH, Haid-und-Neu-Straße 7, 76131 Karlsruhe, Germany

³Institute of Microstructure Technology (IMT), Karlsruhe Institute of Technology (KIT),
Hermann-von-Helmholtz-Platz 1, 76344 Eggenstein-Leopoldshafen, Germany

⁴MultiLane Inc., Houmal Technology Park, Lebanon

⁵Institute of Radio Frequency Engineering and Electronics (IHE), Karlsruhe Institute of Technology (KIT),
Engesserstr. 5, 76131 Karlsruhe, Germany

⁶Institute of Organic Chemistry (IOC), Karlsruhe Institute of Technology (KIT), Kaiserstr. 12, 76131
Karlsruhe, Germany

⁷Institute of Biological and Chemical Systems – Functional Molecular Systems (IBCS-FMS), Kaiserstr. 12,
76131 Karlsruhe, Germany

⁸christian.koos@kit.edu

[†]Contributed equally

*stefan.singer@kit.edu

Abstract: We demonstrate an optically packaged silicon-organic hybrid (SOH) Mach-Zehnder modulator that can be directly driven by industry-standard CMOS SerDes chips without any additional RF amplifiers. In a proof-of-concept experiment we show signaling at PAM4 data rates of 112 Gbit/s with ultra-low peak-to-peak drive-voltage swings down to 265 mV_{pp}. We believe that our work marks an important step towards radically simplified linear-drive optical transceivers with unprecedented efficiency that can rely on sub-500 mV_{pp} CMOS-level drive signals without any additional amplification.

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1. Introduction

Optical transceivers with greatly improved scalability in terms of data rate, integration density, and cost are key to overcome existing and looming interconnect bottlenecks in hyper-scale data-centers and artificial-intelligence (AI) clusters [1]. In this context, linear-drive (LD) optical transmission using technically simple intensity modulation and direct detection (IM/DD) schemes is a promising concept, which eliminates digital signal processing (DSP) in the optical transceiver and instead relies on the serializer/deserializer (SerDes) blocks of the electrical payload integrated circuit (IC). This approach does not only decrease complexity and cost of optical transceivers but also leads to reduced power dissipation and smaller form factor. Clearly, the performance of linear-drive transceivers strongly depends on the underlying electro-optic modulators [2,3], which must combine compactness and amenability to cost-efficient mass production with the ability to operate with standard DSP schemes as offered by SerDes blocks of widely used CMOS ICs. Moreover, compatibility with low drive-voltage swings of 1 V_{pp} or less is desirable to ideally alleviate the need for additional RF drive amplifiers in the electrical signal chain, thereby further reducing the power dissipation. Optical transmission with sub-1 V_{pp} drive voltage

swings has previously been demonstrated using different modulator concepts [4–12]. However, these experiments either relied on rather bulky devices with lengths of the order of 10 mm [8,9], offered only limited IM/DD data rates of 25 Gbit/s or less using simple on-off-keying (OOK) as a modulation format [4,12], were performed with rather complex offline DSP schemes while relying on electrical arbitrary-waveform generators (AWG) for synthesizing the drive signals [5–11], or required fiber amplifiers to boost the optical power levels [4–12]. Linear-drive optical transmission at technically relevant data rates using compact modulators without optical amplifiers hence still remains to be shown.

In this paper we demonstrate amplifier-less linear-drive PAM4 transmission using a CMOS SerDes to directly operate an optically packaged Mach-Zehnder modulator (MZM) [13]. The MZM relies on the silicon-organic hybrid (SOH) integration concept [14] and can be driven by the output signal of an industry-standard 56 GBd PAM4 CMOS SerDes without additional RF amplifiers, offering peak-to-peak drive voltage swings of only 265 mV_{pp} at the electrodes of the MZM. We do not use any optical amplifiers after the modulator and rely on a standard photodiode (PD) with a transimpedance amplifier (TIA) at the receiver. In our proof-of-concept experiments, we demonstrate 112 Gbit/s PAM4 transmission with bit-error ratios (BER) below the KP4 limit at sub-1 mW optical receive powers. To the best of our knowledge, our experiments represent the first demonstration of amplifier-less linear-drive PAM4 transmission relying on a standard CMOS SerDes to directly drive an MZM. Moreover, we demonstrate the first MZM-based transmission link relying on a sub-1 V_{pp} drive-voltage swing without any optical amplification. We believe that our work marks an important step towards radically simplified linear-drive optical transceivers that can simply rely on sub-500 mV_{pp} input signals provided by standard CMOS circuits, without the need for additional DSP or driver electronics.

2. Transceiver scaling and the concept of DSP-less linear-drive optics

Net data rates offered by optical transceivers increased relentlessly over the last 25 years, from 1 Gbit/s (1 G) ethernet interfaces in the 1990s to 1.6 Tbit/s (1.6 T), that are currently being discussed, while 3.2 Tbit/s are being prepared. However, this increase in data rate was also accompanied by a steady increase in power consumption and size of optical transceivers. Figure 1 outlines these trends, showing a timeline of selected ethernet standards and associated form-factors of pluggable transceivers (PT). The dimensions and power ratings provided for the various pluggables in Fig. 1 are extracted from the respective standards [15–21], some of which have been revised several times after initial publication. Depending on the vendor and implemented scheme, the actual power dissipation can be lower than the rating of the respective form factor. Unless noted otherwise, the data rates provided are net data rates, not including the overhead for forward error correction (FEC). Despite the various simplifications, the graph gives a good impression of transceiver evolution. In 1998, the 1 Gbit/s ethernet standard was finalized [22], and the “Gigabit Interface Connector (GBIC)” PT introduced in 1998 supported this data rate in a rather bulky module featuring a volume of 21 cm³ with a power consumption of up to 1.5W[15]. In 2001, the small form-factor pluggable (SFP, not shown) was introduced, reducing the volume by a factor of three to 7 cm³ [16]. With 10 Gbit/s ethernet standardized in 2002[23], SFP was extended to SFP+ in 2006 to be capable of handling the increased data rates, and a maximum power consumption of 2 W was specified [17]. In 2015, 100 Gbit/s ethernet was standardized [24] and the quad small-form-factor pluggable (QSFP) as standardized in 2006 was now more widely adopted. A QSFP28 module supports four optical lanes with a line rate of 28Gbit/s per lane, and the permissible maximum power consumption of a class-8 module amounts to 10 W within a volume of 18 cm³ [18]. By doubling the number of electrical interfaces, the QSFP double-density (QSFP-DD) standard, which was originally introduced in 2014, could support net data rates of up to 400 Gbit/s as needed for the 400 G ethernet standard released in 2017 [25]. The increased data rate again came at the cost of an increased power consumption and size – a QSFP-DD PT can consume up to 25 W and occupy a volume of 21 cm³ [19]. By doubling the number of electrical

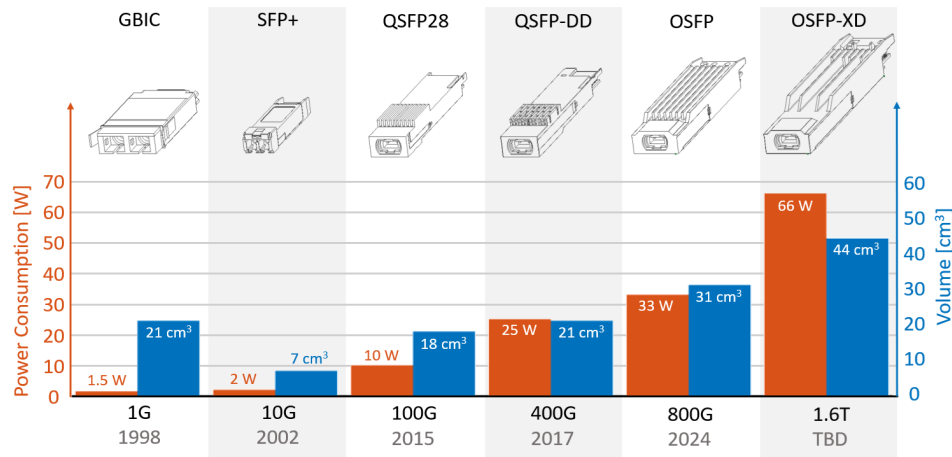


Fig. 1. Timeline of selected ethernet standards and associated pluggable transceiver form factors. The Gigabit Interface connector (GBIC) was introduced in 1998 and supports data rates of 1 Gbit/s while consuming up to 1.5 W of power and occupying a volume of 21 cm³ [15]. The small-form-factor pluggable (SFP, not shown) was standardized in 2001, offering data rates of 2.5 Gbit/s in a volume of 7 cm³ [16]. In 2002, a standard for 10 Gbit/s was introduced [23], leading to the extension of SFP to SFP + in 2006 [17]. SFP + allows for a power dissipation of up to 2 W, while the volume is identical to that of SFP. The next significant step in line rate followed in 2015 with 100 Gbit/s ethernet [24]. The quad small-form-factor pluggable (QSFP28) as standardized in 2006 is capable of supporting this data rate by providing four 28 Gbit/s optical lanes each for transmitting and receiving data. QSFP28 is rated for power consumptions of up to 10 W, and the volume amounts to 18 cm³ [18]. By doubling the number of electronic interfaces, the QSFP double density (QSFP-DD) form-factor, standardized in 2014, can support net data rates of up to 400 Gbit/s as defined in the associated ethernet standard [25], while only slightly increasing the volume. The power dissipation of QSFP-DD modules can be up to 25 W, and the volume amounts to 21 cm³ [19]. In 2024 the 800 Gbit/s ethernet standard was finalized [26]. An appropriate pluggable transceiver for this data rate is the octal small form-factor pluggable (OSFP), introduced in 2017 [20]. The power dissipation of this module can be up to 33 W, and the volume of 31 cm³ is again increased compared to the QSFP-DD. A standard for 1.6 Tbit/s is still being defined, but first 1.6 Tbit/s pluggable transceivers based on the OSFP extra-density (OSFP-XD) form factor are already available. An OSFP-XD pluggable transceiver features a volume of 44 cm³ and is rated for power dissipations of up to 66 W [21]. The illustration clearly indicates a trend towards steadily increasing module sizes and power dissipations, which cannot be sustained in the future – in particular in view of ethernet standards for data rates of 3.2 Tbit/s and more that are already being discussed.

traces again, the octal small form-factor pluggable (OSFP), standardized in 2021, supported net data rates of up to 800 Gbit/s as outlined in the 800 G ethernet standard in 2024 [26]. The maximum power dissipation of an OSFP module is rated to be 33 W, and the volume amounts to 31 cm³ as of the latest revision of the standard in 2022 [20]. For 1.6 T modules, standards are currently in the process of being finalized, relying on the OSFP extra-density (OSFP-XD) as one feasible form-factor [27,28]. OSFP-XD is designed to accommodate 16 optical and electrical inputs and outputs, occupying a volume of 44 cm³ and permitting a power dissipation of up to 66 W [21].

Overall, Fig. 1 clearly indicates a trend towards steadily increasing module sizes and power dissipations, which cannot be sustained in the future – in particular in view of ethernet standards for interfaces rates of 3.2 Tbit/s and more that are already being prepared [27]. As a consequence,

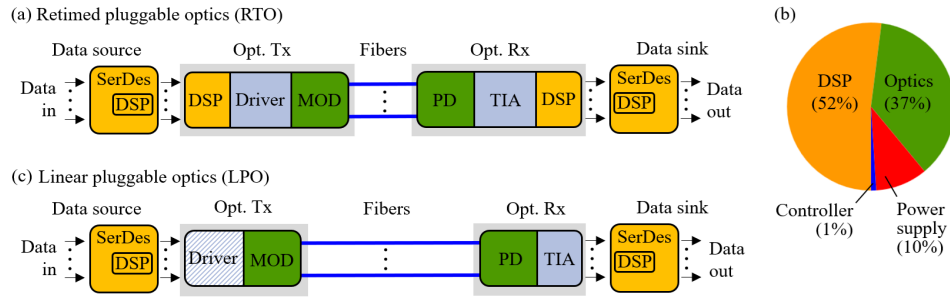


Fig. 2. Conceptual illustration and comparison of a communication link between a digital data source (Data in) and a corresponding data sink (Data out) using conventional retimed optical transceivers, often called retimed pluggable optics (RTO), and transceivers based on linear-drive operation, often referred to as linear pluggable optics (LPO). (a) RTO link architecture. The data source contains a serializer/de-serializer (SerDes) that generates a series of amplitude-modulated electrical data signals, which are sent to the optical transmitter via on-board electrical links. Impairments associated with these links are partially pre-compensated by basic digital signal processing (DSP) blocks in the SerDes, and partially post-compensated by a corresponding DSP block in the optical transmitter (Tx). In addition, the DSP block in the optical transmitter pre-compensates for impairments of the subsequent optical link, generating appropriate drive signals for the subsequent modulators (MOD). These drive signals are boosted by an RF driver (Driver) before being fed to the electrodes of the Tx modulators. This results in corresponding optical signals, which are transmitted over optical fibers and then converted back to the electrical domain using photodiodes (PD) and associated transimpedance amplifiers (TIA) at the optical receiver (Rx). The Rx contains another DSP block, which compensates for remaining impairments of the optical link and at the same time introduces a predistortion to partially compensate for the impairments of the electrical on-board links towards the data sink. The data sink contains again a receiver SerDes with a DSP block that compensates for the remaining impairments of the on-board electrical feed lines. (b) The RTO scheme depicted in (a) is very flexible but suffers from high power dissipation of the DSP blocks in the optical Tx and Rx. The pie chart shows a breakdown of the power dissipation associated with the optical transmitter and receiver (Opt. Tx/Opt. Rx) in (a). DSP blocks in the optical Tx and Rx consume the majority of the overall power, while optical components such as lasers, modulators, and photodetectors consume only about 37% [32]. LPO schemes greatly remove complexity and power dissipation by removing DSP from the optical Tx and Rx. (c) Link architecture of an LPO communication link, omitting DSP blocks in the optical Tx and the optical Rx. If a highly efficient modulator with low drive voltages is used at the Tx side, technically simple CMOS drivers instead of more power-hungry silicon-germanium (SiGe) BiCMOS amplifiers might be sufficient, reducing power consumption further. If the CMOS output signals of the SerDes is sufficient, the RF driver might not be needed at all.

novel approaches that radically simplify the architecture of optical transceivers are becoming imperative. This has led to the concept of linear pluggable optics (LPO) and related linear-drive transceiver schemes, that completely avoid digital signal processing (DSP) in the optical transceiver and that may replace conventional so-called retimed optics (RTO) with internal DSP blocks that are prevailing today. Figure 2 shows a conceptual illustration and comparison of IM/DD communication links utilizing conventional DSP-supported RTO transceivers and LPO-based schemes. In both cases, data from a digital source (Data in) such as a switch application-specific integrated circuit (ASIC), a processor, or a high-speed memory block is fed to a serializer/deserializer (SerDes) unit, which generates a series of amplitude-modulated electrical data signals that are sent to the optical transceiver via an array of parallel electrical high-speed

lanes on a printed circuit board. The source SerDes also contains DSP blocks that partially pre-compensate the impairments of the electrical on-board links. For conventional RTO schemes, Fig. 2(a), these electrical signals are received by another DSP block in the optical transmitter (Tx), which does not only compensate for remaining impairments of the electrical on-board links but also pre-compensates for impairments of the subsequent optical link. This Tx DSP block generates modulator (MOD) drive signals, which are then boosted to the necessary peak-to-peak drive-voltage swings of typically $4 V_{pp}$ [29–31] by an RF drive amplifier (Driver). The optical signals generated by the modulator are transmitted over optical fibers and then converted back to the electrical domain using photodiodes (PD) and associated transimpedance amplifiers (TIA). The output signals of the TIAs are fed to a receiver (Rx) DSP block, which forwards the data to the sink. The Rx DSP block compensates for remaining impairments of the optical link and at the same time introduces a predistortion to partially compensate for the impairments of the electrical on-board links to the data sink. The data sink contains again a receiver SerDes with a DSP block that finally compensates for the remaining impairments of the on-board lines. Clearly, this scheme is very flexible and allows compensating for imperfections of the electrical and optical links on various levels, but the power dissipation of the additional DSP blocks in the optical Tx and Rx is significant. Figure 2(b) shows an exemplary breakdown of the power dissipation of an optical 400 G RTO transceiver, indicating that more than half of the power is consumed by the DSP blocks [32]. LPO can significantly reduce the complexity and power dissipation by omitting DSP blocks in the optical transceivers and by relying on the basic DSP of the SerDes at the data sink and the data source instead, see Fig. 2(c). This scheme holds promise to pave the path towards higher integration density, lower latency, and reduced module costs, all of which are key to implement future communication networks that offer unprecedented bandwidth-scalability [33], e.g., in next-generation AI clusters. In the following, we will report on our proof-of-concept experiments of LPO transmission using highly efficient silicon-organic hybrid (SOH) modulators. Note that these devices may be operated at peak-to-peak drive-voltage swings of a few hundred millivolts [34], which might allow to downsize or even omit dedicated drive amplifiers in the optical Tx. Specifically, these drive amplifiers often rely on power-hungry silicon-germanium (SiGe) BiCMOS technology, and omitting them or replacing them by much more efficient CMOS devices might allow to considerably reduce transceiver power dissipation and complexity.

3. Silicon organic hybrid (SOH) modulator assembly for linear-drive transmission

The electro-optical modulators used in our experiments rely on the concept of silicon-organic-hybrid (SOH) integration [14]. This approach combines the intrinsic scalability advantages of the silicon photonic (SiPh) platform with the ultra-high electro-optic activities of optimized organic electro-optic (EO) materials, for which in-device Pockels coefficients of up to 390 pm/V have been obtained through theory-guided molecular design [6,14]. Besides outstanding efficiency, SOH Mach-Zehnder modulators (MZM) have been shown in previous experiments and associated publications to exhibit low insertion loss [34] and good thermal [5,14,35,36] as well as photochemical [37] stability. Our sample chip carries an array of eight MZM, each of which is optically connected to input and output fibers using 3D-printed waveguide structures, so-called photonic wire bonds (PWB) [38,39]. The die was fabricated as part of an 8-inch wafer through a service offered by Advanced Micro Foundry Pte Ltd. (AMF, Singapore). AMF's platform offers not only a comprehensive portfolio of SiPh devices but also comprises Si_3N_4 waveguides at a different levels of the vertical chip layer stack, which can be used for structures with improved power handling, waveguide crossings, and advanced mode-field converters. Figure 3 shows a top view of the assembly, and Inset (1) gives a more detailed view of the PWB, see Appendix A for a more detailed description of the packaging process. Electrical radio-frequency (RF) drive signals are fed to the device under test (DUT) using ground-signal-ground (GSG) probes at the lower

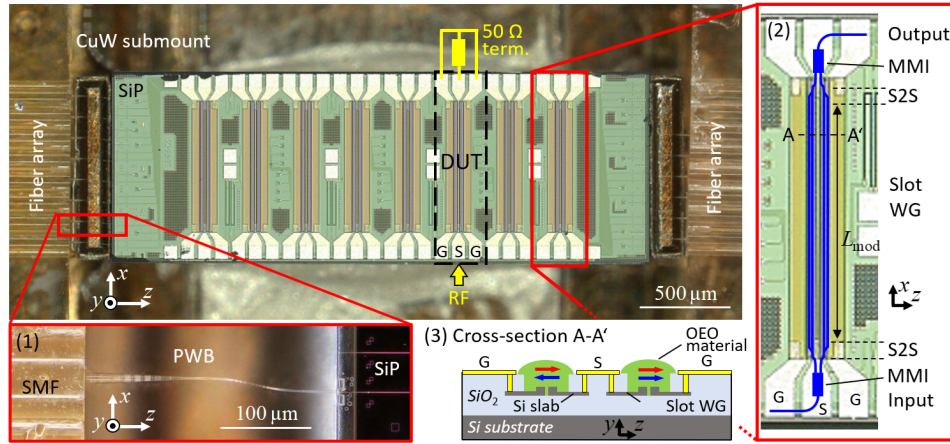


Fig. 3. Microscope image of an optically packaged SOH MZM chip containing an array of eight devices. The die and the associated arrays of input/output fibers are glued on a common CuW submount, and fiber-chip connections are established by photonic wire bonds (PWB) that can compensate for alignment inaccuracies between the fiber arrays and the MZM die, see Inset (1). The RF drive signal is fed to the ground-signal-ground (GSG) coplanar RF transmission line of the respective device under test (DUT) from the lower die edge using an RF probe and propagates through the DUT along with the optical signal. At the other end of the transmission line, the device is terminated by second RF probe that is connected to a $50\ \Omega$ load. Inset (2) shows a more detailed layout of a single MZM, where the optical path is indicated by blue lines. Light is equally split into two paths by a multi-mode interference (MMI) coupler and then coupled to slot waveguides (WG) using strip-to-slot converters (S2S). After passing through the 1 mm-long slot WG, the light enters another S2S converter and is then routed to a second MMI coupler. A cross-section of the modulator along line A-A' is shown in Inset (3). The optical signal is guided by the slot waveguide that is filled with the organic EO material. The rails of the waveguide are connected to the metal GSG transmission line via doped silicon slabs and metal vias. During an initial one-time poling process, the chromophores in the organic EO material are aligned in both slots (red arrows). During operation, the electric field of the RF drive signal is oriented in parallel and antiparallel direction to the poling direction in the right and the left slot of the MZM, respectively, see blue arrows. This leads to efficient push-pull operation.

chip edge in Fig. 3. At the upper edge of the chip, the MZM transmission line is electrically terminated using another RF probe connected to a $50\ \Omega$ load. A more detailed micrograph of a single SOH MZM is shown in Inset (2) of Fig. 3, where the optical path is indicated by artificially added blue lines. Incoming light is equally split into two paths by a multi-mode interference (MMI) coupler. In each path, a strip-to-slot (S2S) converter couples the light from a strip to a slot waveguide (slot width approx. 140 nm, height 220 nm), that is embedded into the organic EO material, see Inset (3) for a cross-section of the device along the line A-A' indicated in Inset (2). The rails of the slot waveguide are connected to the GSG transmission line via doped silicon slabs and metal vias [34,40]. At the other end of the slot WG, another set of S2S converters couples the light back to strip WG that are then combined in a second MMI coupler. For the optically packaged SOH MZM, we measure an optical fiber-to-fiber insertion loss of 10.8 dB, while the optical on-chip excess loss is 4.2 dB. Both numbers leave significant room for further reduction, see Appendix B for a more detailed discussion of the various loss contributions.

Note that the MZM arms were deliberately designed to have a length difference of $\Delta L = 50\ \mu\text{m}$ (not shown), leading to a periodic transmission spectrum with a free spectral range of approximately

12 nm. For such imbalanced MZM, the insertion loss and the extinction ratio can be easily extracted from the optical transmission spectra that are routinely taken as part of wafer-level test procedures. In contrast to that, balanced MZM would need a DC voltage to be applied to the device via a separate electrical probe, and this DC voltage would need to be swept as well. This would complicate wafer-level characterization of the devices considerably. As a side effect, the operation point of unbalanced MZM can simply be set via the wavelength of the carrier.

The organic EO materials can be deposited onto the slot waveguides on wafer scale using, e.g., ink-jet printing or automated dispensing techniques, see Section 5 for a more detailed discussion. After deposition, the chromophores in the organic EO material in both slots need to be aligned in a one-time poling process to establish macroscopic electro-optic activity. In this step, we keep the signal port open and apply the poling voltage U_{pol} between the two floating ground ports while the modulator is heated to the glass transition temperature T_g of the organic EO material. The poling direction is hence the same in both slots of the modulators, indicated by red arrows in Inset (3) of Fig. 3. In contrast to that, operating the device relies on a travelling-wave GSG transmission line, where the drive voltage U_d is fed to the center signal line, see Inset (3) of Fig. 3. This leads to RF field lines that point from the signal line to the two outer ground lines, and that thus have opposite directions in the two slots of the modulator, see blue arrows in Inset (3) of Fig. 3. Poling direction and applied field are hence parallel in the right and anti-parallel in the left slot waveguide, leading to phase shifts in the two MZM arms that are equal in magnitude but opposite in sign. This leads to efficient push-pull operation of the device. Note that the physical picture used in Inset (3) is simplified. In fact, poling is rather a stochastic process, which slightly changes the average orientation of all chromophores but usually does not lead to a complete alignment of all molecules as suggested by the red arrows of Fig. 3, Inset (3). This can be quantified via the so-called average acentric order parameter $AOP = \langle \cos^3(\theta) \rangle$, where θ is the randomly distributed angle between the direction of the poling field and the dipole axis of a chromophore, and where $\langle \cdot \rangle$ denotes the ensemble average over all chromophores in the material. For poled organic EO materials, the AOP is in many cases still smaller than 0.5 [41,42]. After poling of our 1 mm-long device, we measure a π -voltage of 1 V, corresponding to a π -voltage-length product of $U_\pi L = 1$ Vmm. This figure can be further improved – π -voltage-length products as low as $U_\pi L = 0.32$ Vmm have been demonstrated in previous work [6].

One of the key parameters of SOH slot-waveguide modulators is the width of the slot, which for our device was chosen to 140 nm as a good tradeoff between fabrication effort and device performance. From a theoretical point of view, one would expect SOH devices to get more and more efficient with decreasing slot widths, since the effective electrode spacing is reduced. However, there are practical limitations related, e.g., to the resolution of the underlying lithography tools and/or the achievable aspect ratios of the slots. The standard slot width of our devices was deliberately chosen slightly above the limit of the underlying fabrication process to leave some margin for variations without sacrificing yield. Another aspect that needs to be taken into account at narrow slot widths is the fact that efficient poling may be impeded by the tendency of elongated electro-optic chromophores to align in parallel to the slot sidewall rather than perpendicular to it. Such sidewall effects, however, should only become relevant for slot widths below 80 nm [42].

4. Linear-drive transmission experiment

The experimental setup for our linear-drive optical transmission experiment is illustrated in Fig. 4(a). To drive our SOH MZM, we use a bit-error-ratio tester (BERT, ML4039E, MultiLane Inc., Houmal Technology Park, Lebanon) containing a pseudo-random-binary-sequence (PRBS) generator as a data source and a subsequent industry-standard CMOS serializer/deserializer (SerDes) chip which can be operated at PAM4 symbol rates of up to 56 GBd. The CMOS SerDes inside the BERT is directly connected to the corresponding input and output ports, without any amplifiers or other additional electronics interposed. Since our MZM requires only a single-ended

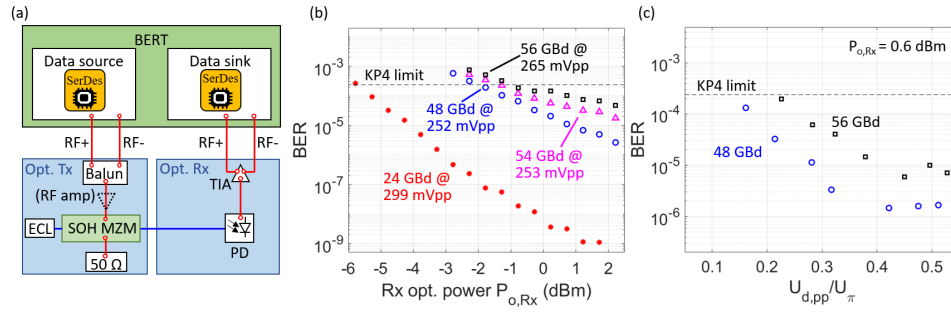


Fig. 4. (a) Schematic of our linear-drive transmission experiment with optional RF amplifier (RF amp). Two electrical payload ICs, that are part of a commercial bit-error rate tester (BERT; MultiLane Inc., Model ML4039E), act as data source and sink. The payload ICs contain only a CMOS serializer/deserializer (SerDes) block with the corresponding standard DSP routines, but no additional RF amplifiers. Since our MZM requires a single-ended drive signal, we use a balancing unit (Balun) to combine the differential output of the SerDes into one single-ended RF port. The resulting signal is either fed directly to the SOH MZM or boosted by an additional RF amplifier (RF amp, SHF 827, 16 dBm saturation output power, 11 dB gain). The MZM is terminated with a 50 Ω load. The optical carrier (1550 nm) is derived from an external-cavity laser (ECL) and fed to the MZM. The ECL, MZM, balun, and the optional RF amplifier form the optical transmitter engine (Opt. Tx). The optical output signal is then sent through a single-mode fiber and detected by an optical receiver (Opt. Rx) module. This module contains a high-speed photodiode (PD) and a subsequent transimpedance amplifier (TIA) with differential outputs, which are directly connected to the input of the Rx SerDes. (b) Measured bit error ratio (BER) vs. received optical power at the input of the opt. Rx for operation of the MZM without RF amplifier. The optical power was swept for different four-level pulse amplitude modulator (PAM4) symbol rates. At a symbol rate of 54 GBd, the BER is below the KP4 limit (black dashed line) for received optical powers above -1.3 dBm. (c) Bit error ratio (BER) vs. normalized peak-to-peak drive voltage $U_{d,pp}/U_{\pi}$ for operation with the additional RF amplifier after the balun. The received optical power is set to 0.6 dBm, and the modulator drive voltage $U_{d,pp}$ is swept with the help of the RF amp. It can be observed that the BER is minimal for a ratio of $U_{d,pp}/U_{\pi} = 0.45$, which is consistent with other transmission experiments [44,45].

drive signal, we include a balancing unit (balun) to combine the two differential outputs of the SerDes into a single-ended MZM drive signal, featuring peak-to-peak voltage swings $U_{d,pp}$ of between 250 mV_{pp} and 300 mV_{pp}, depending on the symbol rate. Note that the balun can be omitted when adopting a differential electrode design for the SOH MZM, as previously shown for plasmonic-organic hybrid (POH) devices [7]. In the subsequent experiments, this signal is either used ‘as is’ to directly drive the MZM, or it is boosted by an RF amplifier (RF amp) prior to being fed to the MZM. The MZM transmission line is terminated with a 50 Ω load. The device is fed by an external-cavity laser (ECL) source which provides a carrier around 1550 nm and which is connected to the MZM by a single-mode fiber (SMF) and a polarization controller (not shown). The wavelength is chosen such that the imbalanced MZM is operated at the quadrature point. The modulated output signal is sent through an SMF and detected by an opto-electronic receiver module (Opt. Rx, MultiLane Inc.), which contains a high-speed photodiode (PD) as well as a transimpedance amplifier (TIA) supplying a differential output signal. The TIA output signal is then directly fed to the receiver SerDes CMOS chip acting as a data sink. For the SOH MZM, the optical fiber-to-fiber insertion loss still amounts to 10.8 dB, while the optical on-chip excess loss is 4.2 dB, both numbers leaving room for further reduction. A more detailed breakdown of the various loss contributions can be found in Appendix B.

In a first set of experiments, we omitted the RF drive amplifier and simply relied on the drive signals supplied by the CMOS SerDes of our BERT. For PAM4 symbol rates of 24 GBd, 48 GBd, 54 GBd, and 56 GBd, we measure peak-to-peak drive voltage swings $U_{d,pp}$ of 299 mV_{pp}, 253 mV_{pp}, 252 mV_{pp} and 265 mV_{pp}, respectively, at the output of the balun, using a high-speed real-time oscilloscope. These signals are fed to the optically packaged MZM using an RF probe, and the received optical power is swept by adjusting the power of the ECL while measuring the bit-error ratio (BER). In all transmission experiments, we use Gray-coded PAM4 signals with a PRBS of length $2^{11}-1$ bits. For the Tx predistortion, we stick to the factory settings of the BERT with three filter taps provided. At the Rx, two equalizers, a multipath-interference canceller, and a reflection canceller are used. All DSP is provided by the SerDes.

In a first step of our signaling experiments, we sweep the optical power of the ECL in the range 8.0 . . 16.3 dBm and measure the associated Rx input power along with the BER. The results are depicted in Fig. 4(b), where the KP4 forward-error-correction (FEC) limit of 2.4×10^{-4} is indicated by the horizontal black dashed line. To benchmark our results, we compare them to the requirements of the IEEE 802.3bs 400GBASE-DR4 [25] standard for short-reach (500 m) ethernet transceivers, transmitting overall data rates of 400 Gbit/s over four optical lanes. According to this standard, each optical lane should operate at a symbol rate of 53 GBd PAM4 with a BER below the KP4 limit, for received optical powers between -5.9 dBm and 4 dBm. As shown in Fig. 4(b), our amplifier-less linear-drive link operates below the KP4 limit for a symbol rate of 54 GBd for received optical powers above -1.3 dBm, such that the requirements of the 400GBASE-DR4 standard are not yet fulfilled but should be within reach. Note also that this standard is defined for the O-band, whereas our current device is still operating in the C-band. As explained in Section 5 below, we expect similar or even slightly better performance for SOH O-band devices [43].

In a second set of experiments, we investigate the impact of higher drive voltages on the BER. To this end, we include an 11 dB RF amplifier (RF amp) at the RF input of the MZM, see Fig. 4(a), and sweep the PAM4 peak-to-peak drive-voltage swing $U_{d,pp}$ at the MZM input for two symbol rates of 48 GBd and 56 GBd at a fixed received optical power of 0.6 dBm. The BER vs. the peak-to-peak drive-voltage swing $U_{d,pp}$, normalized to the MZM π -voltage of $U_{\pi} = 1$ V, is shown in Fig. 4(c). Note that the drive voltage $U_{d,pp} = 265$ mV_{pp}, that is effective when operating the modulator without the RF amplifier, corresponds to only approximately 27% of the π -voltage $U_{\pi} \approx 1$ V. On the positive side, this rather small modulation depth leads to a very linear dependence of the MZM power transmission change on the electrical drive signal. The trade-off, however, is that the spacing of the PAM4 power levels at the MZM output is strongly reduced as compared to the case of a ‘full’ modulation with a peak-to-peak drive-voltage swing that corresponds to the π -voltage $U_{\pi} = 1$ V of the device. This can be observed in Fig. 4(c) – the increase of the drive voltage leads to a decrease of the BER of the resulting optical signal up to the point where nonlinear distortions associated with the cos-type power transfer function of the MZM becomes problematic. We find an optimum $U_{d,pp}/U_{\pi}$ ratio of about 45%, where the PAM4 signal levels are already spaced decently, while the penalty from nonlinear distortions is still acceptable. This observation is in line with other transmission experiments [44,45]. Note that the π -voltage of our device might be reduced by using more efficient materials, which would further increase the performance for amplifier-less operation. Specifically, using more efficient EO materials, π -voltages of $U_{\pi} \approx 320$ mV have been reported for the same device length [6]. In this case, the drive-voltage swing of 265 mV_{pp} provided by the CMOS SerDes without RF drive amplifier would already correspond to more than 50% of the MZM π -voltage, thereby greatly improving the transmission performance and staying below the KP4 FEC limit even without a dedicated RF drive amplifier.

Clearly, while these proof-of-concept experiments demonstrate the fundamental viability of SOH MZM to serve as EO converters in linear-drive transceivers, a more detailed analysis of the signal quality is needed on the path to industrial deployment. Unfortunately, the BERT

used in our experiment only provides the BER and does not give access to eye diagrams or any other signal quality metrics such as the transmitter dispersion eye closure quaternary (TDECQ). Still, our experiments show that SOH MZM can be operated with ultra-low drive voltages while omitting dedicated DSP in the optical transmitter and receiver and relying entirely on the basic DSP routines offered by industry-standard SerDes chips. We find that using highly efficient organic EO materials might even render RF drive amplifiers unnecessary, allowing operation with sub-volt drive-voltage swings that can be directly generated by a standard CMOS circuit. To the best of our knowledge, our experiments represent the first demonstration of amplifier-less linear-drive PAM4 transmission relying on a simple CMOS SerDes to directly drive the MZM. While our experiments represent a first proof of concept, industry-standard performance levels, e.g., with respect to signal quality and power levels are within reach.

5. Future prospects of SOH Mach-Zehnder modulators

The future prospects of using SOH Mach-Zehnder modulators (MZM) in large-scale industrial applications do not only depend on the performance advantages that the technology can offer in comparison to competing modulator platforms, but also on additional aspects such as the amenability to efficient wafer-level mass production, the potential for co-integration with existing silicon photonic circuitry, or questions related to the lifetime and long-term stability of SOH devices. In this section, we benchmark SOH devices with respect to competing technologies and discuss a series of aspects that are specific to the SOH platform such as scalable wafer-level processing, long-term stability, or the ability to address other wavelength ranges such as the telecommunication O-band.

5.1. Benchmarking with respect to other modulator technologies

To benchmark silicon-organic hybrid (SOH) modulator technology to other state-of-the-art platforms, we first compare the π -voltage-length product $U_{\pi}L$ and the π -voltage-loss product $aU_{\pi}L$ [14], see Fig. 5(a). These device-level performance parameters quantify the trade-offs between low operation voltage, small footprint, and low optical loss, all of which are key to linear-drive transceivers. The π -voltage-loss product $aU_{\pi}L$ (unit: VdB) corresponds to the product of the π -voltage-length product $U_{\pi}L$ (unit: Vmm) and the optical propagation loss of the MZM phase shifters (unit: dB/m) and indicates the insertion loss in dB for a device with a π -voltage of 1 V, see [14] for details. We find that SOH MZM, indicated by red stars in Fig. 5(a), offer ultra-low $U_{\pi}L$ products down to 0.46 Vmm [43], only outperformed by plasmonic-organic hybrid (POH) devices, see orange triangles at the right-hand side of Fig. 5(a). However, POH MZM suffer from significant optical loss, leading to rather large π -voltage-loss products $aU_{\pi}L$ in excess of 10 VdB, which make the devices not well suited for linear-drive operation at sub-1 V drive signals without optical amplifiers. In contrast to that, lithium-niobate-on-insulator (LNOI) [46–48] and recently demonstrated lithium-tantalate-on-insulator (LTOI) MZM [49], indicated by black open and filled diamonds in Fig. 5(a), can offer low optical loss with π -voltage-loss products down to $aU_{\pi}L = 0.44$ VdB for LNOI, but suffer from rather large π -voltage-length products $U_{\pi}L$ of typically 20 Vmm or more [47,48]. This leads to rather large device lengths, typically on the centimeter scale. Note that the π -voltage-length product of LNOI MZM can be reduced to approximately 10 Vmm by introducing indium tin oxide (ITO) into the electrode structure, but this comes at the price of an increased optical loss with π -voltage-loss products $aU_{\pi}L$ in excess of 5 VdB [46]. In comparison to conventional silicon photonic (SiPh, blue squares), to InP-based MZM (green crosses), or to devices based on epitaxially grown barium titanate (BTO, violet circles), SOH MZM offer higher efficiency, quantified by lower π -voltage-length products $U_{\pi}L$, while the optical losses are comparable or even lower.

Other key performance metrics of MZM include the bandwidth or, equivalently, the achievable symbol rates, and the amenability to cost-efficient mass production. With respect to the latter, a

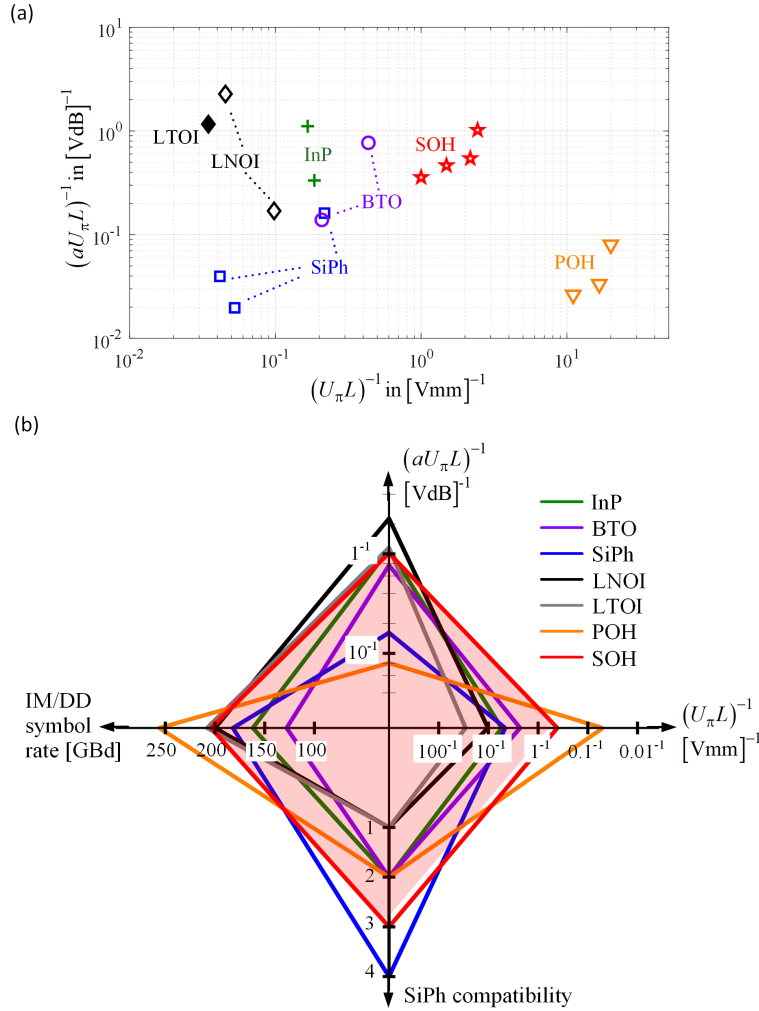


Fig. 5. Benchmarking of different Mach-Zehnder modulator technologies. (a) Comparison of the π -voltage-length product $U_{\pi}L$ and the π -voltage-loss product $aU_{\pi}L$ as key device-level performance parameters of MZM. Note that the axes indicate the inverse values $(U_{\pi}L)^{-1}$ and $(aU_{\pi}L)^{-1}$ such that device performance increases towards the upper right-hand corner of the plot. The data is taken from the references shown in Table 1 below. SOH MZM offer higher efficiency than most competing technologies, quantified by lower π -voltage-length products $U_{\pi}L$, while the optical losses are comparable or even lower. An exception are POH devices, which offer even smaller $U_{\pi}L$ products, but suffer from significant optical loss, quantified by rather large π -voltage-loss product $aU_{\pi}L$ in excess of 10 VdB. LNOI = Lithium-niobate-on-insulator MZM; LTOI = Lithium-tantalate-on-insulator MZM; SiPh = Silicon photonic MZM; InP = Indium-phosphide-based MZM; BTO = MZM based on epitaxially grown barium titanate; SOH = Silicon-organic hybrid MZM; POH = Plasmonic-organic hybrid MZM. (b) Radar chart comparing MZM technologies with respect to four performance metrics: The π -voltage-length product $U_{\pi}L$ and the π -voltage-loss product $aU_{\pi}L$ as device-level performance parameters, the maximum IM/DD symbol rates demonstrated in laboratory signaling experiments, and the compatibility with the SiPh platform, that is key to cost-efficient mass production of optical transceivers. The SiPh compatibility is quantified on a scale from 1 to 4, depending on the technological effort associated with combining the respective modulator technology with advanced SiPh circuitry in a chip-scale package. The radar chart indicates that SOH technology offers a very attractive overall performance profile, including the amenability to wafer-level co-integration with advanced SiPh circuitry without the need for adapting front-end-of-line fabrication processes.

Table 1. Comparison of different Mach-Zehnder modulator (MZM) technologies and associated signaling experiments. The data in Columns 2, 3, 4, and 5 has been used and discussed in Fig. 5 above. Columns 6 indicates the peak-to-peak drive voltage used in the signaling experiment if mentioned in the underlying reference. Columns 7 and 8 indicate the signal source and sink as well as the digital signal processing (DSP) scheme used in the signaling experiment. While many other experiments rely on arbitrary-waveform generators (AWG) and high-speed oscilloscopes (OSC) as signal source and sink in combination with offline DSP at the receiver, our demonstration simply uses an industry-standard CMOS SerDes, thereby demonstrating the ability of SOH devices to enable radically simplified linear-drive transceiver concepts. Similarly, most previous signaling demonstrations required optical amplifiers to overcome device losses, whereas our demonstration was performed without optical amplifier.

Techn.	$U_{\pi}L$	$aU_{\pi}L$	SiPh comp.	IM/DD symbol rate / Mod. format	Peak-to-peak drive voltage	Signal source / sink	DSP scheme	Opt. ampl.	Ref.
SOH	1.00 Vmm	2.80 VdB	3	56 GBd / PAM4	0.27 V _{pp}	CMOS SerDes	CMOS SerDes	No	This work
	0.41 Vmm	0.98 VdB	3	100 GBd / PAM4	1.44 V _{pp}	AWG / Osc.	Offline	Yes	[34]
	0.67 Vmm	2.15 VdB	3	204 GBd / PAM4	-	AWG / Osc.	Offline	Yes	[52]
	0.46 Vmm	1.84 VdB	3	176 GBd / PAM8	0.95 V _{pp}	AWG / Osc.	Offline	Yes	[43]
SiPh	19 Vmm	50.7 VdB	4	182 GBd / OOK	-	AWG / Osc.	Offl. DSP at Tx	Yes	[53]
	4.6 Vmm	6.2 VdB	4	154 GBd / PAM4	2.88 V _{pp}	Not stated	Not stated	Yes	[54]
	24 Vmm	25.2 VdB	4	24 GBd / OOK	2.30 V _{pp}	AWG / Osc.	Offline	Yes	[55]
InP				100 GBd / PAM4					
	5.4 Vmm	3.0 VdB	2	80 GBd / PAM8	2.30 V _{pp}	PPG&DAC / Osc.	Offline	Yes	[56]
	6.0 Vmm	0.9 VdB	2	162 GBd / PAM16	-	AWG / Osc.	Offline	Yes	[57,58]
BTO	2.3 Vmm	1.3 VdB	2	124 GBd / PAM4	4.00 V _{pp}	PPG&DAC / Osc.	No DSP	Yes	[59]
	4.8 Vmm	7.2 VdB	2	128 GBd / OOK	0.83 V _{pp}	AWG / Osc.	Offline	Yes	[60]
				112 GBd / PAM6	3.30 V _{pp}				
				110 GBd / PAM8	3.70 V _{pp}				
LNOI	10.2 Vmm	5.9 VdB	1	112 GBd / OOK	-	AWG / Osc.	Offline	No	[46]
				112 GBd / PAM4					
	22 Vmm	0.44 VdB	1	100 GBd / OOK	-	AWG / Osc.	Offline	Yes	[47]
	28.8 Vmm	-	1	56 GBd / PAM4					
				200 GBd / PAM4	<1.00 V _{pp}	AWG / Osc.	Offline	Yes	[48]
				172 GBd / PAM6					
				180 GBd / PAM8					
LTOI	28.8 Vmm	0.86 VdB	1	208 GBd / OOK	-	AWG / Osc.	Offline	Yes	[49]
				200 GBd / PAM4					
				176 GBd / PAM8					
POH	0.05 Vmm	12.5 VdB	2	-	-	-	-	Yes	[42]
	0.09 Vmm	38 VdB	2	256 GBd / OOK	-	AWG / Osc.	Offline	Yes	[61]
				200 GBd / PAM4					
				176 GBd / PAM8					
	0.06 Vmm	30 VdB	2	-	-	-	-	Yes	[62]

key aspect is the ability to co-integrate MZM with silicon photonic (SiPh) circuitry, which has become the mainstay of optical transceivers [50]. In this respect, the SOH concept stands out due to its good compatibility with the SiPh platform, relying fully on back-end-of-line hybridization, see Section ‘Special aspects of SOH modulator technology’ below for a more detailed discussion. To provide a simple overview of these multi-dimensional performance metrics, we summarize them in a radar chart, i.e., a two-dimensional chart of four metrics represented on different axes starting from the same central point, see Fig. 5(b). The four metrics comprise the π -voltage-length and π -voltage-loss products, $U_{\pi}L$ and $aU_{\pi}L$, respectively, the highest IM/DD symbol rates demonstrated in transmission experiments (‘IM/DD symbol rate’), and the compatibility with the silicon photonic platform (‘SiPh compatibility’). The SiPh compatibility is quantified on a scale from 1 to 4, where four points are assigned to SiPh MZM themselves, whereas SOH is assigned three points due to the additional effort of back-end-of-line hybridization. InP, BTO-based MZM, and POH MZM might also be integrated on the SiPh platform, e.g., through heterogeneous integration concepts [51], but this requires changes of front-end-of-line processes, which leads to an increased technological complexity. We hence assigned two points to these approaches. For LNOI and LTOI MZM, the challenge is the large device lengths, which are often on the centimeter scale – a significant mismatch to mm-scale SiPh on-chip circuitry. Co-integration of such devices with SiPh circuitry is therefore likely to rely on rather costly co-packaging of different chiplets – we assigned one point to this approach. The radar chart indicates that SOH technology offers a very attractive and highly competitive overall performance profile, that may be outperformed by competing approaches only in specific performance metrics.

Table 1 finally summarizes the data used in Fig. 5(a) and (b) and provides the associated references. Regarding data-transmission demonstrations, Table 1 also specifies the peak-to-peak drive voltage, Column 6, as well as the type of signal source and the digital signal processing (DSP) scheme, Columns 7 and 8, and further indicates whether or not an optical amplifier was used to boost the power of the data signal, Column 9. Note that our current work not only relies on an industry-standard CMOS SerDes but is also one of the few signaling experiments that did not rely on an optical amplifier – a key aspect when it comes to compact and efficient linear-drive transceiver links.

5.2. Special aspects of SOH modulator technology

As discussed in the previous section, SOH technology stands out due to a unique combination of efficiency, low on-chip losses, high modulation speed, and the amenability to co-integration with advanced SiPh circuitry, making the platform highly attractive for scalable linear-drive optical transceivers for hyper-scale data centers and AI clusters. This moves a series of platform-specific aspects into the focus of the discussion, comprising scalable wafer-level processing, long-term device stability and operation at elevated temperatures, as well as operation in the telecommunication O-band. We address these aspects in the following.

- **Scalable wafer-level processing:** As a key advantage, SOH technology can purely rely on back-end-of-line processing of SiPh wafers without any need for changing front-end-of-line fabrication steps. The back-end-of-line processing requires three steps: Deposition of the organic EO material, hermetic encapsulation, which is key to long-term photochemical stability [37], as well as poling of the material to establish macroscopic EO activity. For the first two steps, working on the wafer level is advantageous to scalable processing. More specifically, material deposition can rely on fast ink-jet printing techniques, for which cycle times of less than 1 s per MZM can realistically be achieved. This would permit a throughput of the order of 10 million devices per year for a single printer. Encapsulation can rely on wafer-level thin-film deposition techniques [37], which allow for highly efficient parallel processing and which should hence not represent a scalability bottleneck. Scaling of the poling process is slightly more challenging, since it simultaneously requires heating

to the glass transition temperature T_g of the EO material and applying a poling voltage to the device. Also here, processing should ideally be performed on the wafer level, but this requires not only tight control of the temperature distribution across the wafer, but also techniques to apply the poling voltage simultaneously to all devices. As an alternative, SOH devices could be poled on a chip-by-chip base, which would simplify the process while reducing the throughput. These questions and the associated wafer-level processing techniques are subject of ongoing research.

- Thermal stability and operation range:** Another important aspect of SOH MZM is related to the compatibility of these devices with elevated temperatures. In the past, discussions about temperature-sensitivity of SOH devices have rather revolved around the aspect of long-term stability of the underlying organic electro-optic (EO) materials under the influence of heat, e.g., in the context of high-temperature storage. We have previously demonstrated two ways to achieve long-term storage stability of organic EO materials, using either materials with high glass-transition temperatures [5] that offer long-term stability at typical temperatures of 85°C, or relying on cross-linking of the organic matrix after poling, through which operation temperatures of 120°C can be reached [63]. With these stability levels being compatible with widely used industry standards [64], we feel that long-term temperature stability of SOH devices is not a fundamental problem any more. Another aspect is the question to which extent the performance of the SOH MZM degrades when operating the device at elevated temperatures. This was investigated in a previous study by operating an SOH MZM at a temperature of 80°C [65]. We observed a slight increase of the π -voltage U_π that was still typical for the organic EO materials used at the time [66] and that is meanwhile not a problem any more [5]. Beyond that, no temperature-induced performance degradations could be observed, and we may hence assume that SOH devices based on state-of-the-art materials should maintain their performance also at elevated operating temperatures. More recently, we have also explored the other end of the temperature scale and operated SOH MZM at cryogenic temperatures of a few Kelvin [67]. Also here, we found that the π -voltages of the devices measured at cryogenic temperatures were still essentially the same as the room-temperature U_π . Based on these findings, we believe that SOH devices do not only offer long-term storage stability at elevated temperatures, but can also maintain their performance over a large thermal range, from cryogenic temperatures up to temperatures of 85°C or more.
- Photochemical stability and encapsulation:** For practical applications, SOH MZM must be able to handle high on-chip power levels. In the context of our BER measurements, we performed a sweep of the MZM input power levels to vary the received optical power, see Fig. 4(b) and the associated discussion above. In these experiments, the power in the fiber leading towards the SOH MZM was varied in the range 8.0 . . . 16.3 dBm, which lead to power levels in the range 0.2 . . . 8.5 dBm at the input of each of the two on-chip MZM phase shifters, see Table 2 in Appendix B for the insertion losses of the fiber-chip interface and the various on-chip components. Note that SOH MZM have been shown to exhibit long-term photo-stability for more than 2000h using even slightly higher power levels up to 10 dBm per slot-waveguide phase shifter, provided that suitable encapsulation techniques are used to prevent in-diffusion of ambient oxygen [37].
- Operation in the telecommunication O-band:** Organic electro-optic (EO) materials designed for O-band operation are in many cases slightly more lossy while exhibiting also higher EO activity than their C-band counterparts [68,69]. For a given π -voltage, the higher material efficiency allows to reduce the devices length, which counteracts the increase of the overall device loss while leading to higher modulation bandwidth due to reduced RF losses. We have demonstrated the viability and performance of O-band SOH MZM

in another work [43], where we reached π -voltage-length products of $U_{\pi}L = 0.46$ Vmm along with phase-shifter propagation losses of approximately 4 dB/mm, leading to a π -voltage-loss product of $aU_{\pi}L = 1.84$ VdB. These numbers are even slightly better than those of the C-band device used in the current paper, for which the π -voltage-length product of $U_{\pi}L = 1.0$ Vmm and the phase-shifter propagation losses of approximately 2.8 dB/mm led to a π -voltage-loss product of $aU_{\pi}L = 2.8$ VdB. Note that the O-band device had a length of only 0.5 mm with a π -voltage of 0.92 V and was shown to support PAM4 O-band transmission up to 192 GBd (line rate 384 Gbit/s).

- Co-integration with state-of-the-art silicon photonic and silicon nitride (Si_3N_4) circuitry:** The SOH process technology is amenable to co-integration with SiPh circuitry, allowing to combine the advantages of SOH modulators with the vast portfolio of advanced SiPh devices such passive components or photodetectors. With silicon nitride (Si_3N_4) becoming a more and more important material in the context of SiPh circuits, the question arises to which extent the SOH concept could actually benefit from this material system. Note that the devices used in our experiments rely on an industry-standard silicon photonics platform (Advanced Micro Foundry, AMF, Singapore), that already includes Si_3N_4 waveguides at a different level of the layer stack [70]. These Si_3N_4 waveguides lend themselves to implementing low-loss waveguide crossings and passive devices that can handle much higher power levels than their SiPh counterparts, which additionally increases the attractiveness of the SOH integration concept. Note, however, that building organics-based phase shifters by directly using Si_3N_4 slot waveguides instead of their conventional SiPh counterpart is yet another challenge. Unlike the weakly doped silicon rails, see Inset (3) of Fig. 3, Si_3N_4 is an insulator, and potential Si_3N_4 rails cannot serve as closely spaced modulator electrodes. Moreover, the refractive index of Si_3N_4 and most organic EO materials are fairly similar, such that highly confined guidance of light like in a conventional SiPh slot waveguide might be hard to achieve. Hence, while silicon-nitride-organic-hybrid (SiNOH) lasers sources have been demonstrated in the context of point-of-care diagnostics [71,72], organics-based modulators on the silicon nitride platform would certainly represent another important breakthrough.

6. Summary

We have demonstrated amplifier-less linear-drive PAM4 signaling at peak-to-peak drive-voltage swings of 265 mV_{pp} and line rates of up to 112 Gbit/s using an optically packaged SOH MZM. The drive signal is provided by an industry-standard CMOS SerDes chip offering peak-to-peak voltage swings between 250 mV_{pp} and 300 mV_{pp}. These signals are either coupled directly to the MZM or via an additional RF drive amplifier. Our scheme does not require any optical amplifiers and shows the potential to even obviate the RF drive amplifier. We analyze the power-dependent BER performance and find the requirements of the IEEE 802.3bs 400GBASE-DR4 standard to be within reach. We believe that our work marks an important step towards radically simplified linear-drive optical transceivers that can simply rely on sub-500 mV_{pp} CMOS-level input signals, offering unprecedented efficiency and scalability.

Appendix A – Optical packaging of modulator dies

In our experiment, light is coupled from standard single-mode fibers (SMF) to the modulator die via edge couplers and photonic wire bonds (PWB) [38,73,74], see Inset (1) of Fig. 3. We use fiber arrays rather than bare fibers for easier handling and fix them next to the silicon photonic die on a common CuW submount using UV glue. Photonic wire bonding relaxes the alignment tolerances of the SMF to the input- and output waveguide of the modulator, where the distance between the fiber and chip facets (z -direction) should be approximately 250 μm and the lateral offsets (x - and

y-direction) should be smaller than 30 μm . This allows for mounting of the chips and fiber arrays using passive alignment based on standard machine-vision techniques. The PWB are printed on an industry-grade multi-photon lithography system (Sonata 1000, Vanguard Automation GmbH, Karlsruhe, Germany) using a dedicated photoresist (VanCore B, Vanguard Automation GmbH, Germany; refractive index $n = 1.53$ at $\lambda = 1.55 \mu\text{m}$). The machine automatically detects the fiber and chip interfaces and calculates the optimum shape of the PWB, taking into account a series of parameters such as offsets and tilt angles of the fiber and the chip, and different mode-field diameters at the coupling interfaces. For the MZM chip used in our demonstration experiments, we estimated insertion losses of approximately 3.3 dB per PWB, see Appendix B, clearly leaving room for further improvement. Specifically, optimization of the 3D-nanoprinting process can reduce the losses of a single-mode PWB-based fiber-chip interface to less than 2 dB while offering a yield in excess of 95%, specified after 2000h of damp-heat testing, see [75] for details. Interestingly, the results presented in [75] were obtained for exactly the same silicon photonic platform that was also used in our experiments, such that the same performance should be reachable with our SOH devices. Note also that PWBs are often designed to exhibit a slightly elliptical cross section [76], which leads to two non-degenerate orthogonally polarized fundamental modes, thus rendering the PWB-structure polarization-maintaining. Targeted design of the PWB cross section can even be exploited to implement polarization rotators and splitters [39].

Another interesting question related to the photonic wire bonding technique has to do with the scalability potential and the industrial maturity of the overall approach. Using optimized writing processes along with suitable lithography hardware [77], PWB exposure times can be brought down to approximately 30 s per structure, making the technique suitable for mass production. It should still be noted that the SOH approach is compatible with a wide range of packaging and assembly processes and is by no means limited to photonic wire bonding only. In that respect, the scalability of the packaging process is not a specific aspect of SOH technology, but rather a challenge for low-cost transceivers in general. We are also using SOH devices in combination with other packaging approaches, relying, e.g., on so-called facet-attached microlenses (FaML) [78,79]. This concept also relies on 3D-printed coupling structures and can leverage the same lithography hardware, while offering a slightly different workflow, that might be advantageous in some cases.

Appendix B – Breakdown of optical loss contributions

Our current MZM assembly has a fiber-to-fiber insertion loss of 10.8 dB, which already permits transmission without optical amplifier while still leaving room for further improvement. A breakdown of the optical loss contributions of the fiber-chip interfaces and of the various on-chip building blocks of our current assembly is given in Table 2. The relevant loss parameters for our current implementation are shown in Columns 2 and 5. The optical propagation loss of the slot waveguide in the phase-shifter sections of the MZM and of the strip waveguides that connect the modulator to the chip facets were characterized via cutback measurements of dedicated test structures of different lengths, produced in the same fabrication run as the chip used in our assembly. For these cutback measurements, the slot waveguides were embedded into an organic electro-optic (EO) material that is equivalent to the one used in our devices under test, but that was not poled. From these cutback measurements, we find propagation losses of 2.8 dB/mm for the slot waveguide, which corresponds to an insertion loss of 2.8 dB for the 1 mm-long slot-waveguides used in our MZM. Note that this number is consistent with the previously reported sub-1 dB loss of a 280 μm -long SOH phase shifter [34]. Note also that we did not apply any encapsulation to the organic EO material for simplicity – we found in previous experiments that neither the poling nor the encapsulation has any impact on the propagation losses of the underlying slot-waveguide phase shifters. To extract the losses of

the multi-mode interference (MMI) couplers and the strip-to-slot (S2S) converters, we also used cutback measurements with cascades of such devices. The MMI structures of our devices are fully covered by silicon dioxide (SiO_2 , refractive index $n = 1.44$ at $\lambda = 1.55 \mu\text{m}$), whereas the strip-to-slot converters are covered by the organic EO material (refractive index $n = 1.8$ at $\lambda = 1.55 \mu\text{m}$) with a transition to the silicon-dioxide cladding in the strip-waveguide section, where the impact is much lower than in the slot waveguide. Note that these transitions were also incorporated into the cutback test structures such that the measured strip-to-slot converter losses already include the losses associated with the transition of the cladding materials. We also investigated the back reflections of the cladding-material transitions using numerical simulations (Lumerical FDTD solver, Ansys Inc., Canonsburg, United States). We find S_{11} -parameters of the interface well below -40 dB , which should not represent a problem in applications. This is confirmed by our cutback measurements of strip-to-slot converter cascades - in these experiments, back reflections originating from the cladding-material transitions would have resulted in standing waves and hence in a significant wavelength dependence of the transmission of the cascade, which we did not observe in our measurements. The low back-reflections of the cladding-material transitions originates from the fact that the majority of the mode field is confined within the 500 nm -wide silicon waveguide core such that the impact of the discontinuity in the cladding is rather low.

Table 2. Breakdown of optical loss contributions of the fiber-chip interface and of the various on-chip building blocks of our device. For the current device, the total fiber-to-fiber loss of the fully assembled device amounts to 10.8 dB , see Columns 2 and 5. Further optimization of device designs, materials, and fabrication processes could reduce the various loss contributions – this is indicated Columns 3 and 6, leading to a total fiber-to-fiber loss between 5 and 6 dB . WG = waveguide, MMI = multi-mode interference, S2S = strip-to-slot converter, PWB = photonic wire bond.

Element	Propagation loss / excess loss per element		Length / number of elements	Total loss	
	Current	Optimized		Current	Optimized
Strip WG	0.14 dB/mm	0.1 dB/mm	4.8 mm	0.7 dB	0.5 dB
Slot WG	2.8 dB/mm	1.5 dB/mm	1 mm	2.8 dB	1.5 dB
MMI coupler	0.25 dB	0.1 dB	2 MMI	0.5 dB	0.2 dB
S2S converter	0.1 dB	0.1 dB	2 S2S	0.2 dB	0.2 dB
PWB	3.3 dB	1.5 dB	2 PWB	6.6 dB	3.0 dB
Sum				10.8 dB	5.4 dB

Note that most of the losses stated in Column 5 of Table 2 can be reduced further – the corresponding values are indicated in Columns 3 and 6 and could lead to a total fiber-to-fiber loss between 5 and 6 dB . More specifically, optimized geometries of MMI couplers can lower the excess loss to values of less than 0.1 dB [80,81], while the loss of the phase shifter section can be reduced by using optimized organic EO materials with lower absorption losses and/or by optimized etching processes that lead to lower sidewall roughness and correspondingly reduced scattering loss. Another option is to use shorter phase shifter sections that offer smaller overall scattering loss while relying on more efficient EO materials to maintain the π -voltage. The insertion loss of the PWB can be improved with optimized printing processes, through which losses below 2 dB have previously been shown for the same type of fiber-chip connection [75], see Appendix A above. For the strip-to-slot converter, the current design is already quite close to the expected minimum value [80].

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